
ZOOLOGY

The Growth and Energy Metabolism of *Lymnaea stagnalis* (Lymnaeidae, Gastropoda): I. Early Postlarval Period

A. A. Zotin

Kolzov Institute of Developmental Biology, Russian Academy of Sciences, ul. Vavilova 26, Moscow, 119334 Russia

e-mail: aazotin@mail.ru

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Abstract—The growth and the oxygen consumption rate of *Lymnaea stagnalis* were studied during the first ten weeks after hatching. It is shown that these processes are atypical during early ontogenesis in comparison with adult mollusks. The obtained data on linear (height of shell) and weight growth can be equally well approximated with the von Bertalanffy equation or exponential and power equations. Both linear and weight growth are characterized by an approximately constant specific rate associated with synchronous oscillations of a week period. The oscillations were observed also for the oxygen consumption rate, but of another period (about 2.6 weeks). On average the metabolic rate after the initial triple increase during the first three days remains stable. The power coefficient of the allometric dependence of the total weight on the shell height is significantly less than that of the adult.

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INTRODUCTION

The growth and the change of energy metabolism of adult animals overwhelmingly comply with the same pattern, whereby it's possible to use the same equations for description of these processes in the most diverse of animal species (Zotin, Zotina, 1975, 1993; Mina, Klevezal', 1976; Alimov, 1981; Zotin, Zotina, 1999). If the increase in weight is permanent, the von Bertalanffy equation (Bertalanffy, 1960) is commonly used for the description of the growth. Known as the finite type of growth, i.e., growth that is characterized by maximum sizes with their posterior stabilization and reduction, it could be described by our equation (Zotin, 2006). In the same paper, an equation approximating well the change of the oxygen consumption rate during postlarval (postnatal) ontogenesis of different animals was proposed.

The collected data show that the growth and the energy metabolism pattern peculiar to adults cannot be used for early postlarval ontogenesis (Blaxter, 1969; Ivanova, 1973; Ozernyuk, 1985, 2000a, b; Katoh, 1989; Sokolova, 2000; Maximova et al., 2007; Cook et al., 2007).

The term "early postlarval period" is extremely fuzzy, and it is interpreted in different ways depending on the author's viewpoint and species studied. Therefore, we should be specified what is understood by this term in this work.

It is essential to define the early postlarval period as a specific stage because, after the end of metamorphosis and development of the juvenile animal in general, there is a dramatic change in the environmental parameters. A definite period of time is required for the organism to acclimate to the new conditions of its existence.

Evidently, the beginning of the early postlarval period is the point of passage to adult existence: hatching, if previous development takes place inside of an egg, case, etc.; leaving the maternal organism in the case of viviparity; or the end metamorphosis and passage to conditions of adult existence for animals with free larva. The end of the early postlarval period is the moment of complete acclimation to the new environmental conditions when organism's parameters (weight and metabolic rate) begin to comply with the pattern character significant for adult animals. Certainly, it is understood that environmental parameters remain more or less unchanged.

This work is a continuance of previous study on the growth and energy metabolism of the freshwater gastropod *L. stagnalis* during embryogenesis (Zotin, Kleimenov, 2006), and it is dedicated to study of the weight and oxygen consumption rate change in juvenile *L. stagnalis* during the first 10 weeks after hatching.

MATERIAL AND METHODS

Juvenile *L. stagnalis* L. (Lymnaeidae, Gastropoda) were grown in laboratory conditions. The animals used had hatched as spawn from a previous experiment (Zotin, Kleimenov, 2006). In total, 517 specimens were studied.

Mollusks were kept in the same conditions as the spawn: in previously settled water (3–4 days) under constant temperature 18°C in 100 ml plastic cups. In order to prevent mollusks from escaping from the plastic cups, they were covered with a capron mesh of 0.5 mm. Water

and food were changed two times a week. As food we used white cabbage *Brassica oleracea* L.

The test-sensitivity of the oxygen consumption rate did not permit measurements to be carried out for individual specimens immediately after hatching. Therefore, most juvenile mollusks were cultivated in groups (up to 20 specimens per group). The group contained animals with the same size.

When individual specimens reached a shell size of 2 mm, the groups were separated and then the animals were kept individually. Individual measurements were possible after the test-sensitivity permission.

During first four weeks, measurements of the size (height and diameter of shell), total weight, and the oxygen consumption rate were carried out two times a week, and thereafter, once a week.

Linear sizes were measured using binocular adjustment with an eyepiece micrometer if the shell height was up to 5 mm, otherwise we used a vernier caliper accurate to within 0.1 mm.

Animals with a weight up to 250 mg were weighed using a torsion balance accurate within 1 mg. If the weight of an individual specimen was less than 3 mg, all mollusks from the group were weighed, then the specimen weight was calculated. The total weight of animals of large sizes was measured using a rough balance accurate within 10 mg.

Previously five measurements were carried out in water for *L. stagnalis* from 1 to 7 days after hatching using a polarographic gas analyzer (Zotin, Kleimenov, 2006). The number of analyzed animals was 20–40 specimens. The oxygen consumption rate for juvenile *L. stagnalis* was not revealed in these experiments. Apparently mollusks while underwater use the oxygen that was captured from the atmosphere and kept under the shell surface. In this connection, further measurements of the oxygen consumption rate were carried out using the Warburg method (Umbreit et al., 1951) for animals kept on air under humidity close to 100% and a temperature of 18°C for 2–4 hours. The mass specific oxygen consumption rate was calculated dividing the value of the individual oxygen consumption rate to the weight of this specimen.

The obtained data were fitted with a smoothing cubic spline, after that the weight and the linear growth rates were calculated using the Matlab program (version 7.3.0.267) developed by the MathWorks Inc. (United States). The specific weight growth rate and height specific rate of height growth were calculated dividing the absolute rate to the proper value of the weight and height of the shell obtained after fitting.

For description of growth, most often equations having linear form were used (Zotin, Zotina, 1993):

The von Bertalanffy equation

$$Y = Y_{\infty}(1 - \exp(-k_g(t + t_0)))^u.$$

The linear form of this equation is the Ford–Walford recurrent equation (Ford, 1933; Walford, 1946)

$$Y_{t+1}^{1/u} = cY_t^{1/u} + d, \quad (1)$$

where Y_t is the growth parameter (weight or height of the shell) at the age of t weeks after hatching; Y_{t+1} is the same parameter in the same group (or individual specimens) at the age of $t + 1$ weeks; Y_{∞} is the limit value (asymptote) of the Y parameter if $t \rightarrow \infty$, k_g , t_0 , u are the coefficients; $c = \exp(-k_g)$, $d = (1 - c)Y_{\infty}$.

The power equation

$$Y = (ct + d^{1/u})^u \quad (2)$$

$$(\text{linear form} - Y^{1/u} = ct + d^{1/u}).$$

The exponential equation

$$Y = d \exp(uct) \quad (3)$$

$$(\text{linear form} - \ln Y^{1/u} = ct + \ln d^{1/u}).$$

In the case of linear growth, the constant u was considered equal to 1; in the case of weight growth, it was regarded that $u = k = 2.33$, where k is the exponential coefficient of allometric dependence of the total weight on the shell height (4). Thus, the power equation is equivalent to the equation of the line for the linear growth. The coefficient d could be interpreted as the shell height or mollusk weight during hatching ($t = 0$) in equations (2) and (3).

The allometric equation was used for description of the correlation between the measured parameters:

$$Y = aX^k \quad (4)$$

$$(\text{linear form} - \ln Y = k \ln X + \ln a),$$

Y , X are the measured parameters and a , k are the coefficients.

The acceptability of the linear form of equations is estimated by the nonlinear criterion (Zotin, 2000).

The sinusoid equation was used for the parameter estimation of the oscillation processes:

$$Y = A \sin(2\pi(t + t_0)/P) + Y_{av}, \quad (5)$$

where A is the amplitude, P is the cycle of oscillation, t_0 is the phasephase, and Y_{av} is the average value of the Y -parameter. The approximation was calculated using Matlab (version 7.3.0.267).

RESULTS

Mass mortality of the mollusks was observed immediately after hatching. According to our observations, the root cause of the mortality concerned the migratory activity of juvenile animals: they moved into the air and tried to leave the cup. Many of them “adhered” to the cup wall and died. Mollusks that returned to the aquatic environment were suppressed, and cases of their death were frequent. Of the 517 studied individuals of

Table 1. The growth and the energy metabolism of *L. stagnalis* during early postlarval ontogenesis

<i>t</i> , weeks	Growth				Energy metabolism			
	<i>N</i>	<i>n</i>	<i>M</i> , mg	<i>L</i> , mm	<i>N</i>	<i>n</i>	<i>Q</i> , mcl O ₂ /h	<i>q</i> , ml O ₂ /(h g)
0*	17	843	0.38 ± 0.01	1.25 ± 0.01	17	843	0.035 ± 0.001	0.093 ± 0.002
0.5	12	224	0.43 ± 0.03	1.35 ± 0.03	12	224	0.136 ± 0.017	0.314 ± 0.033
1	13	246	0.53 ± 0.04	1.55 ± 0.03	13	246	0.147 ± 0.010	0.285 ± 0.019
1.5	14	231	0.76 ± 0.04	1.73 ± 0.03	14	231	0.152 ± 0.013	0.201 ± 0.015
2	21	332	0.61 ± 0.05	1.51 ± 0.05	21	332	0.160 ± 0.014	0.271 ± 0.021
2.5	18	177	0.84 ± 0.05	1.80 ± 0.04	17	175	0.171 ± 0.034	0.203 ± 0.029
3	19	219	0.69 ± 0.06	1.68 ± 0.05	19	219	0.139 ± 0.023	0.207 ± 0.022
3.5	8	35	1.05 ± 0.12	1.98 ± 0.09	7	32	0.261 ± 0.079	0.218 ± 0.044
4	13	74	0.86 ± 0.15	1.80 ± 0.08	11	69	0.384 ± 0.081	0.444 ± 0.058
5	8	38	1.23 ± 0.21	2.08 ± 0.17	7	37	0.349 ± 0.074	0.285 ± 0.033
6	18	38	1.34 ± 0.15	2.17 ± 0.09	15	33	0.351 ± 0.058	0.327 ± 0.046
7	11	25	1.70 ± 0.18	2.42 ± 0.10	1	1	0.450	0.366
8	18	25	1.95 ± 0.26	2.53 ± 0.13	18	25	0.599 ± 0.114	0.317 ± 0.046
9	6	6	2.91 ± 0.56	3.02 ± 0.30	5	5	0.542 ± 0.141	0.367 ± 0.208
10	9	9	3.28 ± 0.56	3.14 ± 0.32	6	6	0.687 ± 0.071	0.328 ± 0.055

Annotation: *t* is the time from the moment capsule escape, *N* is the number of measurements; *n* is the number of animals; *M* is the weight; *L* is the shell height; *Q* is the oxygen consumption rate; and *q* = *Q*/*M* is the mass specific rate of oxygen consumption.

* Data were obtained during studies of *L. stagnalis* embryogenesis (Zotin, Kleimenov, 2006).

L. stagnalis, 24 specimens (around 4.5%) lived until 10 weeks of age, and 14 specimens (around 2.5%) reached adulthood.

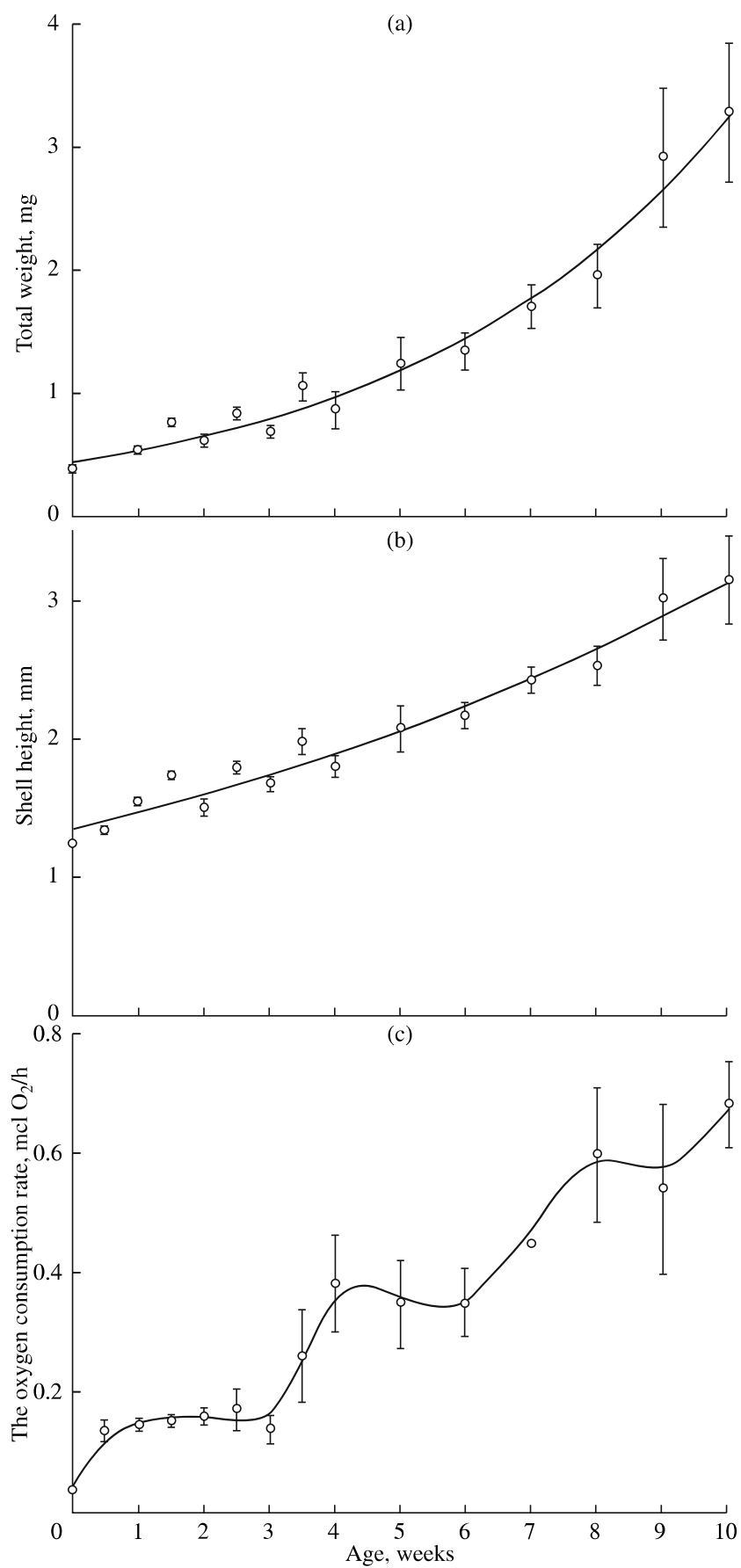
From three weeks of age, individuals with a higher growth rate appeared among the total sample. The weight and shell height of these mollusks was characterized by rapid growth, but the specific growth rate and the oxygen consumption rate continuously declined after reaching the maximum. A constant decrease in the specific growth rate and the oxygen consumption rate is typical for adult animals. Therefore, based on the above definition, it could be considered that the period of the maximum level of these parameters corresponds to the end of early postlarval development.

The length of early postlarval development in *L. stagnalis* varies, and it ranges from 3 to 19 weeks after hatching (at 18°C). Only 36 specimens (around 7%) completed this period. There were 4 specimens (11%) of 3–5 weeks of age in this group; 8 specimens (22%) of 6–8 weeks of age; 19 specimens of 9–11 weeks of age, and 5 specimens of 12–19 weeks of age.

The average experimental data found during early postlarval ontogenesis are shown in Table 1.

At first sight the weight and shell height of mollusks change quite gradual (Figs. 1a, 1b). However, the calculations of the specific growth rate (*dM*/*Mdt*) and the height specific rate of shell height growth (*dL*/*Ldt*) have shown that this is not the case: the growth of mollusks is characterized by variable rate. Synchronous oscillations (Fig. 2) were revealed for both parameters. The average values of these parameters are constant during the whole early postlarval period, and they coincide with the values of the end of the embryonic period.

The approximation of data on growth by different equations showed that all the equations used (the von Bertalanffy equation (1), power equation (2), and exponential equation (3)) described the experimental data satisfactorily. This was confirmed by the nonlinear criterion. The values of equation coefficients are shown in Table 2. The growth curves based on three equations almost coincide (Figs. 1a, 1b). The disparity in the values of functions on the interval 0–10 weeks is no more



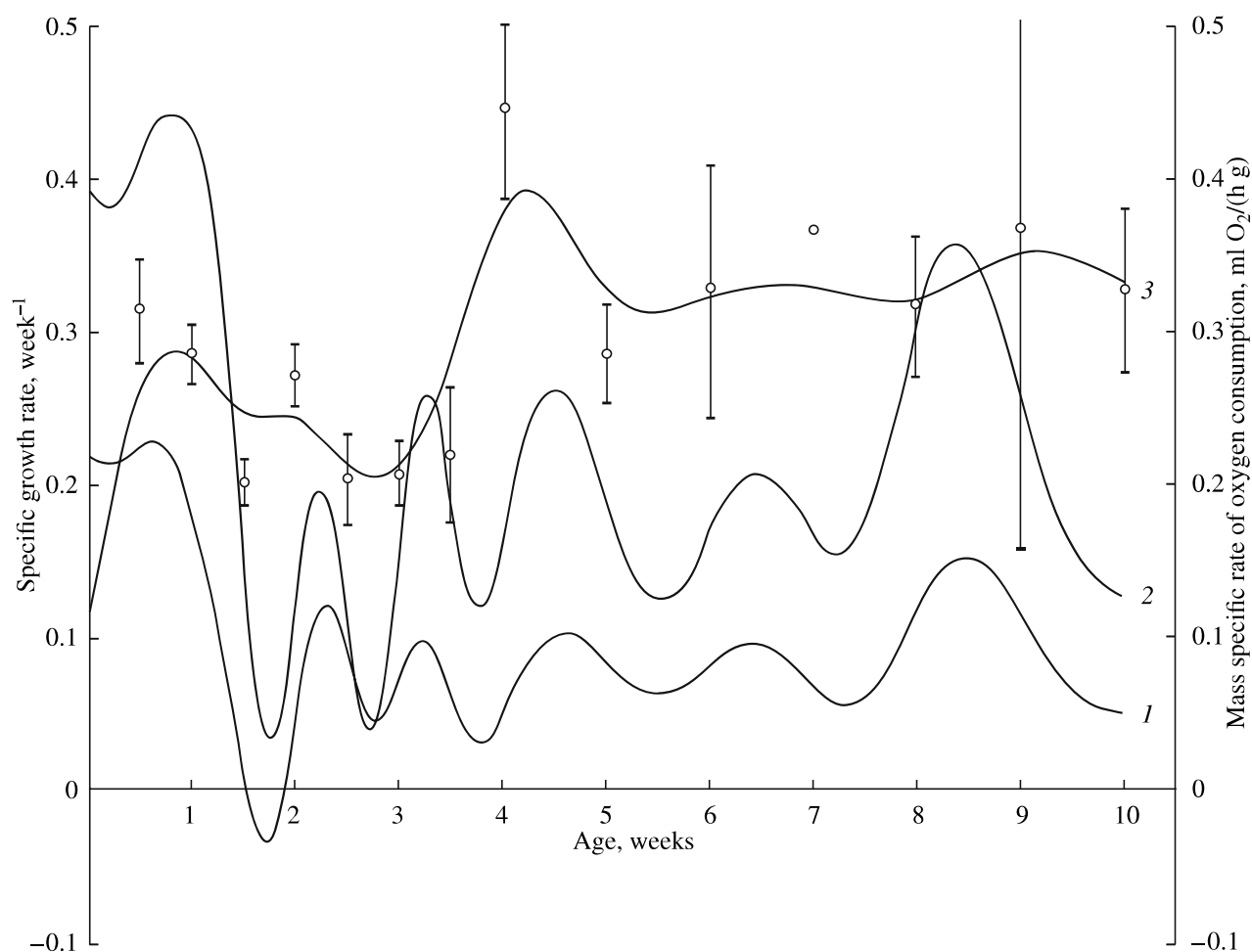


Fig. 2. Change in the specific weight growth rate (1), the height specific rate of height growth (2) (left Y-direction), and the mass specific rate of oxygen consumption (3) (right Y-direction) during early postlarval ontogenesis of *L. stagnalis*. The lines are drawn using smoothing cubic splines. The circles are the experimental data on the mass specific rate of oxygen consumption.

than 0.26 mg for the weight growth and 0.24 mm for the linear growth (shell height).

The oxygen consumption rate (Q) increases during the early postlarval period. The Q change is graduated: increase periods during 1 week change into 2-week constancy periods (Fig. 1c).

Significant oscillations were detected for the mass specific rate of oxygen consumption (q). The age when the oscillations reach the local maximums, and their cycles do not coincide with the corresponding values for dM/Mdt and dL/Ldt . The average q after the initial thrice-repeated increase during the first three days is relatively constant (Fig. 2). Thus, taking into account the increase in the total weight, it could be considered that the intensity of energy metabolism does not depend

on the body weight during early postlarval ontogenesis of *L. stagnalis*.

The values of oscillation parameters fitted by the sinusoid equation are shown in Table 3.

In spite of the irregularity of change in the growth and metabolism parameters, the interdependency between them was very well fitted by the allometric (power) equation; it was confirmed by the nonlinear criterion. The allometric dependences are as follows:

$$M = (0.207 \pm 0.035)L^{(2.33 \pm 0.03)}, \quad (n = 191);$$

$$D = (0.696 \pm 0.004)L^{(0.66 \pm 0.02)}, \quad (n = 182);$$

$$Q = (0.263 \pm 0.062)M^{(0.89 \pm 0.08)}, \quad (n = 170);$$

Fig. 1. Change in the total weight (a), the shell height (b), and the oxygen consumption rate (c) during early postlarval ontogenesis of *L. stagnalis*. Curves a and b are an approximation by equations (1), (2), (3) (the curves of all equations are congruent); c is a smoothing by cubic splines. The circles are the experimental data.

Table 2. Values of equation coefficients after fitting of growth data

	Linear growth		Weight growth	
	<i>c</i>	<i>d</i>	<i>c</i>	<i>d</i>
Ford–Walford equation (1)	0.913 ± 0.042	0.368 ± 0.080	0.949 ± 0.042	0.156 ± 0.041
The power equation (2)	0.167 ± 0.011	1.322 ± 0.054	0.086 ± 0.006	0.395 ± 0.038
The exponential equation (3)	0.076 ± 0.005	1.406 ± 0.034	0.077 ± 0.005	0.454 ± 0.026

Annotation: $u = 1$ for linear growth, $u = 2.33$ for weight growth. Units: weight, mg; shell height, mm; age, weeks. The number of value pairs $Y_{t+1} - Y_t$ for Ford–Walford equation (1) is 103; the number of measurements for other equations is 191. In the von Bertalanffy equation, t_0 is 3.5 weeks for the linear growth, t_0 is 3.3 weeks for the weight growth.

Table 3. The rhythm parameters of specific growth rate and the mass specific rate of oxygen consumption

Parameter	dM/Mdt	dL/Ldt	<i>q</i>
Average value	$0.219 \pm 0.009 \text{ weeks}^{-1}$	$0.094 \pm 0.103 \text{ weeks}^{-1}$	$0.301 \pm 0.005 \text{ ml O}_2/\text{h}$
Amplitude	$0.063 \pm 0.013 \text{ weeks}^{-1}$	$0.044 \pm 0.007 \text{ weeks}^{-1}$	$0.035 \pm 0.007 \text{ ml O}_2/\text{h}$
Cycle	$1.96 \pm 0.04 \text{ weeks}$	$2.00 \pm 0.03 \text{ weeks}$	$2.63 \pm 0.08 \text{ weeks}$
Phase	$0.19 \pm 0.12 \text{ weeks}$	$0.07 \pm 0.09 \text{ weeks}$	$0.81 \pm 0.15 \text{ weeks}$

Note: dM/Mdt is the specific weight growth rate; dL/Ldt is the height specific rate of height growth; and *q* is the mass specific rate of oxygen consumption.

where *M* is the weight (mg); *L* is the shell height (mm); *D* is the shell diameter (mm); *Q* is the oxygen consumption rate (mcl O₂/h); and *n* is the number of measurements.

According to the correlation between the diameter and height of the shell, it is possible to draw the conclusion that since the power coefficient $k = 0.66 < 1$, then the shell tends to growth in height faster than in diameter.

The power coefficient of the *Q*(*M*) correlation is not significantly different from 1; this confirms the conclusion about the independence of the mass specific rate of oxygen consumption on weight during early ontogenesis of *L. stagnalis*.

DISCUSSION

The obtained data give evidence that the patterns of growth and energy balance during early postlarval ontogenesis of *L. stagnalis* are differ from those of adult animals. The specific weight growth rate, height specific rate of height growth, and mass specific rate of

oxygen consumption during this period did not change on average, while a constant decrease in these parameters was revealed during late ontogenesis (Zotin, Zotina, 1975, 1993; Mina, Klevesal', 1976).

References concerning growth during early postlarval (postnatal) ontogenesis are not numerous. However, the available data are sufficient to state the absence of a pattern common to all species for growth and metabolism during this period.

Overall, it is possible to point out the following variants of growth in Gastropoda:

1. Exponential (concave curve): linear and weight growth of freshwater mollusks

Maackia herderiana (Maksimova et al., 2007).

2. Linear (straight line): linear growth of marine (*Cypraea annulus*) and amphibious

(*Littorina saxatilis*) gastropods (Sokolova, 2000).

3. S-shaped: weight growth of marine gastropod *Babylonia areolata* (Kritsanapuntu et al., 2006), *Bucci-*

num undatum (Nasution, Roberts, 2004), *Cypraea annulus* (Kato, 1989).

4. Parabolical (convex curve): linear growth of freshwater (*Potamopyrgus antipodarium*), marine (*Babylonia areolata*) (Kritsanapuntu et al., 2006), and amphibious (*Littorina brevicula*) species (Takada, 2003).

The exponential type of linear size growth during early postlarval ontogenesis is also described in crustaceans (Ivanova, 1973; Cook et al., 2007); exponential, linear, and parabolical, in fish larva (Blaxter, 1969; Ozernyuk, 2005). The growth of fish larva is probably an artifact as the increase in weight a short time after hatching is characterized by watering at least in some species. Actually, the dry weight of fish larva decreases probably as a consequence of starvation because of a shortage of nutrients in the vitellus remains. Only the switch to external nutrition causes the weight to increase (Shatunovskii, 1980).

Only the first three variants of growth indicated above are appropriate (in accordance with nonlinear criterion) for the obtained data. This would mean the equations describing these growth types have the same recurrence formula (1) with different values of angular coefficient c : $c < 1$ for the von Bertalanffy equation, $c = 1$ for the linear equation, and $c > 1$ for the exponential equation. Thus, according to the value of the angular coefficient, it is possible to draw a conclusion about the most suitable type of equation for description of the experimental data. However, in this case it is not possible to do this as the distinction of c from 1 is not significant (Table 2). If we take into consideration that the growth curve is the same as in adults by the von Bertalanffy equation (1), then in this case the growth is also different during early and late postlarval ontogenesis. Indeed, it follows from coefficient values of the Ford-Walford equation (Table 2) that the extreme dimensions of juvenile *L. stagnalis* are 13.7 mg (total weight) and 4.3 mm (shell height). These values are no less than 3 g and 33 mm, respectively, for adult animals.

The constancy of the oxygen consumption rate during early postlarval ontogenesis of *L. stagnalis* also does not agree with the change kinetics in this parameter during late ontogenesis. A decrease in the intensity of energy metabolism with age was achieved in the overwhelming majority of adult animals of different taxa using a great amount of experimental data (Zotin, 1993).

The references concerning change in the mass specific rate of oxygen consumption during early postlarval (postnatal) development of different taxa animals do not permit us to recognize any general patterns: cases of increase in this parameter (Alekseeva, 1985; Radzinskaya, Nikol'skaya, 1986; Ozernyuk, 1985, 1988, 2000a, b; Ozernyuk, Lelyanova, 1985; Alekseeva et al., 1991; Radzinskaya et al., 2003;) and its constancy (Etkin, 1934; Fletcher, Myant, 1978; Radzinskaya et al., 2003) or decrease (Laurence, 1975, 1978;

Forstner et al., 1983; Radzinskaya, Nikol'skaya, 1982; Nikol'skaya et al., 1986; De Silva et al., 1986; Wieser, Forstner, 1986; Kleimenov, 1996; Vladimirova et al., 2000, 2003) have been observed.

The dependence of the total weight (mg) on shell height (mm) of adult *L. stagnalis* according to Alimova and Golikova (1974) is described by the allometric correlation $M = (0.24 \pm 0.02)L^{(2.79 \pm 0.20)}$. Our value of the exponential coefficient is significantly lower ($p < 0.05$). Hence, the allometric growth is different during early and late postlarval ontogenesis.

Rhythmical processes of the specific growth rate and the height specific rate of shell height growth are characterized by a nearly two week period during early postlarval ontogenesis. Rhythms of growth with the same cycle were described for mammals, birds, and amphibian (Kopeç, 1932, 1937; Fedorov, 1947, 1973; Sipachev, 1970). According to these authors, the rhythms are endogenous and independent of periodical processes in the environment.

Furthermore, permanent oscillations are found for the mass specific rate of oxygen consumption. Whereas the rhythms of change of weight and shell height are synchronous, the oscillations of the mass specific rate of oxygen consumption have another period (approximately 2.6 weeks); i.e., the processes of biosynthesis and energy metabolism occur independently of one another. Nevertheless, it is early to draw conclusions about the oscillation processes revealed as long as their cycle is congruent to the period of experimental data organization and the oscillations could be characterized by chance oscillation of the measured parameters. Most likely, more detailed investigations will provide an answer to this question.

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